F1 Podium Prediction – Milestone 1 Report

The table below maps each requirement of the project brief to the section and figures where we address it.

Requirement (project brief)	Section	Figures
EDA before and after cleaning, with plots	2	01–03, 06–07
Sentinel handling, semantic typing, keys, one row per (raceId, driverId), duplicate rule	2	06b
Data-engineering questions Q1–Q3: answers, grain, missingness, cutoff	3	08–10
Selected features, evidence, and why each is valid at the pre-race cutoff	4	11, 13b
Leakage-safe form and standings features	4	—
Train / validation / test split and its justification	5	—
Preprocessing and its effect on the data	5	15
Models (statistical + shallow NN), baseline, how they work, limitations	5, 6	17
PR-AUC, ROC-AUC and F1 on train, validation and unseen test; primary / secondary metric justified	6	18–19b
Preprocessing-order experiments (at least two)	7.1	—
Feature-group ablation	7.3	20
Global XAI: permutation importance and SHAP	8.1, 8.2	21–23
Local XAI: SHAP waterfall and LIME	8.3	24–25
Explanations are not causal (stated explicitly)	8	—
Known limitations	9	—
Inference function on a complete and a partially-missing record	10	—

1. Summary

Our final recommendation is a logistic regression trained on 18 pre-race features. On the held-out 2022–2024 seasons it reaches a PR-AUC of 0.722, a ROC-AUC of 0.937 and an F1 score of 0.673. For comparison, a random model scores a PR-AUC of 0.150 on these seasons and simply ranking drivers by grid slot gives 0.645. The random forest, the two gradient-boosting models and the shallow neural network all land within the uncertainty of the logistic regression, so we kept the simplest model, which is also the best calibrated.

The task is to predict, for every driver on the starting grid, whether that driver finishes on the podium (`positionOrder ⩽ 3`). Only information available after qualifying and before the start of the race may be used.

All steps, plots and numbers in this report come from our run-all notebook (`F1_Podium_Prediction.ipynb`). Figure numbers follow the file names in the notebook's `outputs/figures/` folder.

Item	Result
Modeling table	24,715 driver–race rows, 1,114 races, unique on (raceId, driverId)
Split	train 1950–2019 (22,576 rows) · validation 2020–2021 (780) · test 2022–2024 (1,359)
Primary / secondary metric	PR-AUC / ROC-AUC; F1, precision and recall at a validation-chosen threshold
Best test PR-AUC	logistic regression and random forest, 0.722 each
Strongest feature group	grid (starting position); grid alone loses 0.084–0.100 validation PR-AUC

2. Data, audit and cleaning

After cleaning, the 26,759 raw result rows became 24,715 rows with exactly one row per driver on the grid per race. The raw CSV files were never modified; we compared their MD5 hashes before and after cleaning to confirm this.

We used version 24 of the Kaggle / Ergast *Formula 1 World Championship (1950–2024)* dataset, which has 14 relational tables and 1,125 races. Every file was first loaded as plain text so that the audit sees the values exactly as they are stored. The audit found the following issues:

- **Sentinels.** 30 columns in 6 tables hold the placeholder `\N`, plus 68 empty strings in `qualifying.q2/q3`. Pandas detects 0 missing values in the raw text, so these are hidden missing values *(fig 01)*.
- **Era drift.** Entries per race peaked above 30 in the late 1980s against 20 today, so only 8–12% of rows were podiums in busy eras vs ~15% now *(fig 02)*.
- **`grid = 0` means two things.** Before 1995, 1,334 of 1,537 grid-0 rows are failed qualifiers; from 1995 most are pit-lane starters.
- **Qualifying coverage.** `qualifying.csv` starts in 1994, covers only 6–59% of races in 1996–2002, and is complete from 2003 *(fig 03)*.
- **Inconsistent text.** `"Argentinian "` vs `"Argentine"`, `"United States"` vs `"USA"`, and one HTML code (`Lotus-Pratt & Whitney`).

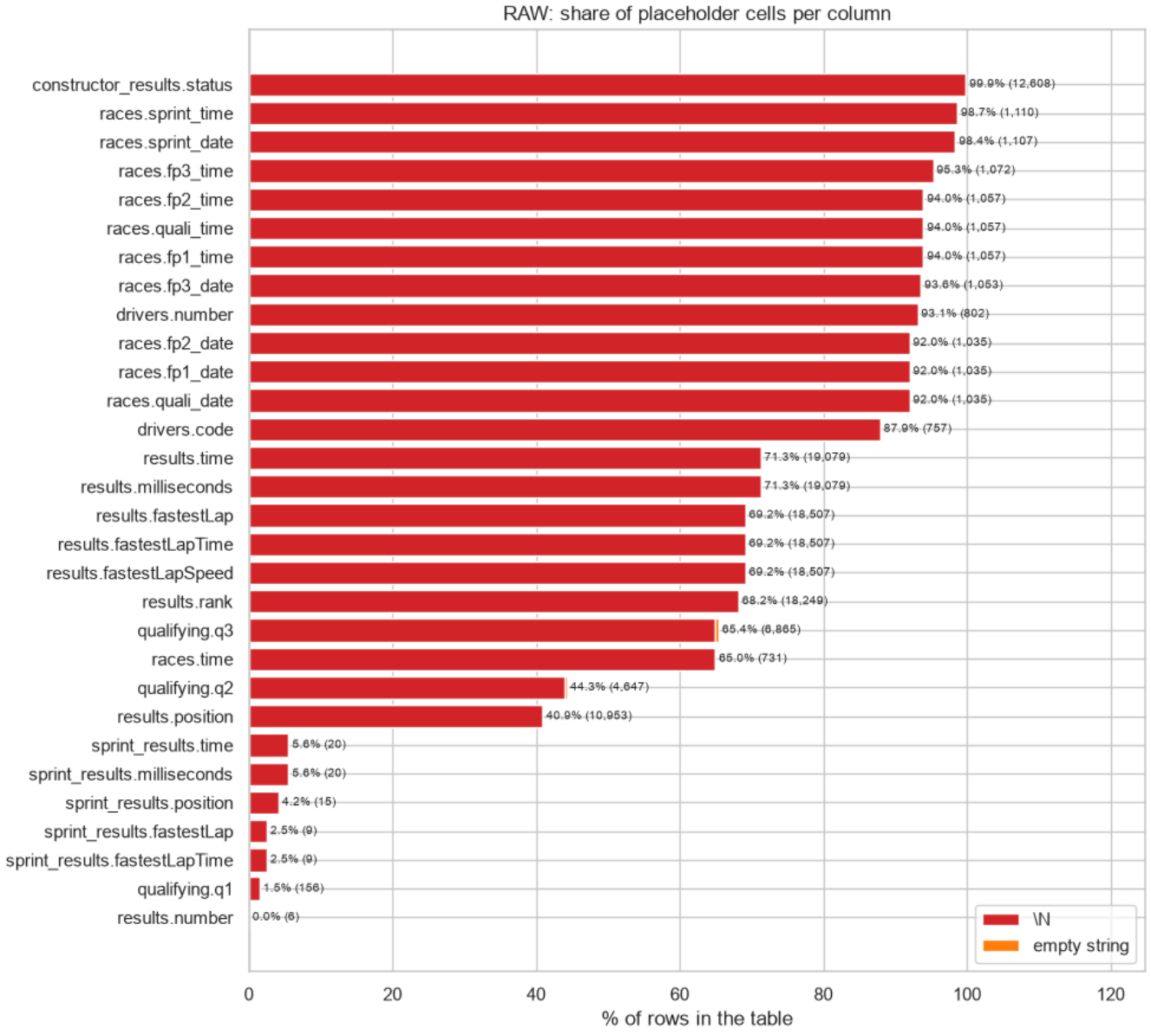


Fig 01. Raw data: share of `\N` and empty-string cells per column.

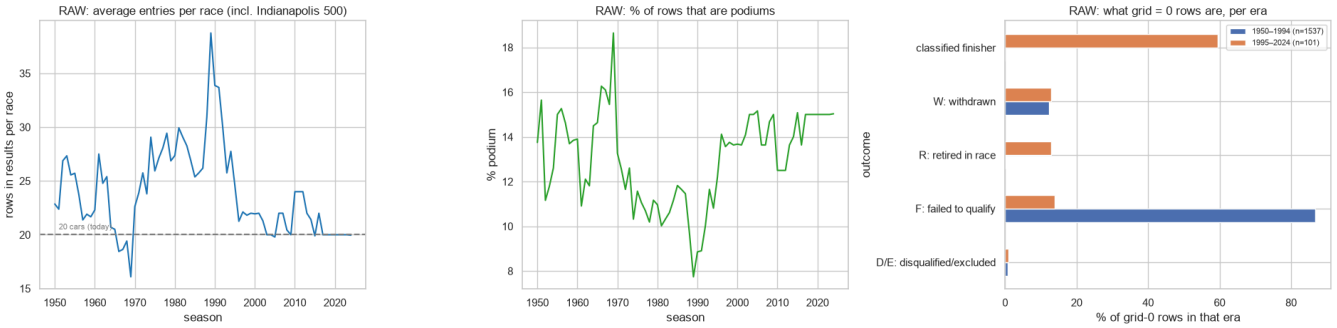


Fig 02. Raw data over 75 seasons: entries per race, podium share of rows, and what `grid = 0` means per era.

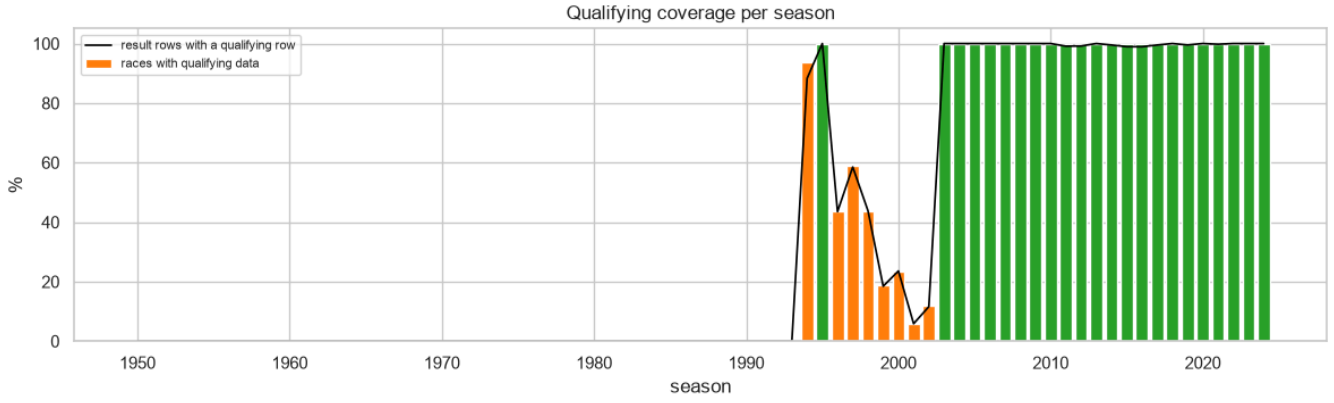


Fig 03. Qualifying coverage per season: none before 1994, partial 1996–2002, complete from 2003.

The cleaning steps and their effect on the data are summarised below.

Step	What was done	Effect
Sentinels	`\N` and `""` read as NaN; a cell-by-cell check confirms NaN ⇔ placeholder in every column	0 disagreeing cells
Semantic typing	dates → `datetime`; lap, qualifying and pit times → seconds; race time from `milliseconds`	18 impossible race times (2024 Belgian GP) set to NaN
Text normalisation	spelling and HTML fixes (D2)	43 → 42 nationalities, 35 → 34 countries
Keys	primary and natural keys unique and non-null; 26 foreign-key links checked	0 orphans
Status categories	139 labels mapped one-to-one to 8 categories (finished, mechanical, accident/driver, …)	used for Q3 and the grid filter

Step	What was done	Effect
Duplicates (shared drives)	85 (raceId, driverId) pairs, 1950–1978; keep the car only that driver is recorded in, else the lowest `resultId` (D4)	91 rows removed
Indianapolis 500	excluded: a separate oval race with 33+ car fields and almost no F1 regulars (D5)	381 rows removed
On the grid at cutoff	failed qualifiers and never-on-grid entries removed; did-not-start cars kept (D6)	1,365 + 207 rows removed
Grid recoding	grid re-ranked 1…N per race; pit-lane starts set to the last slot plus a flag (D7)	0 rows with grid 0
Target	`podium = positionOrder ⩽ 3` (D8)	base rate 13.5%

For the duplicates we compared three rules against two integrity checks: at most three podium rows per race, and one driver per car per race. Keeping each driver's best position left 15 races with more than three podium drivers, keeping the lowest `resultId` left 2, and our chosen rule left only 1. It also keeps the car the driver actually started in, which is the car that the pre-race information (the grid slot) refers to.

Comparing the data before and after cleaning (figs 06, 06b), duplicate driver–race pairs drop from 91 to 0 and rows with grid 0 from 1,638 to 0. The podium base rate rises from 12.7% to 13.5% because drivers who failed to qualify, and so could never podium, are no longer counted. Grid position predicts podiums in every era, and the relationship is stronger in recent seasons.

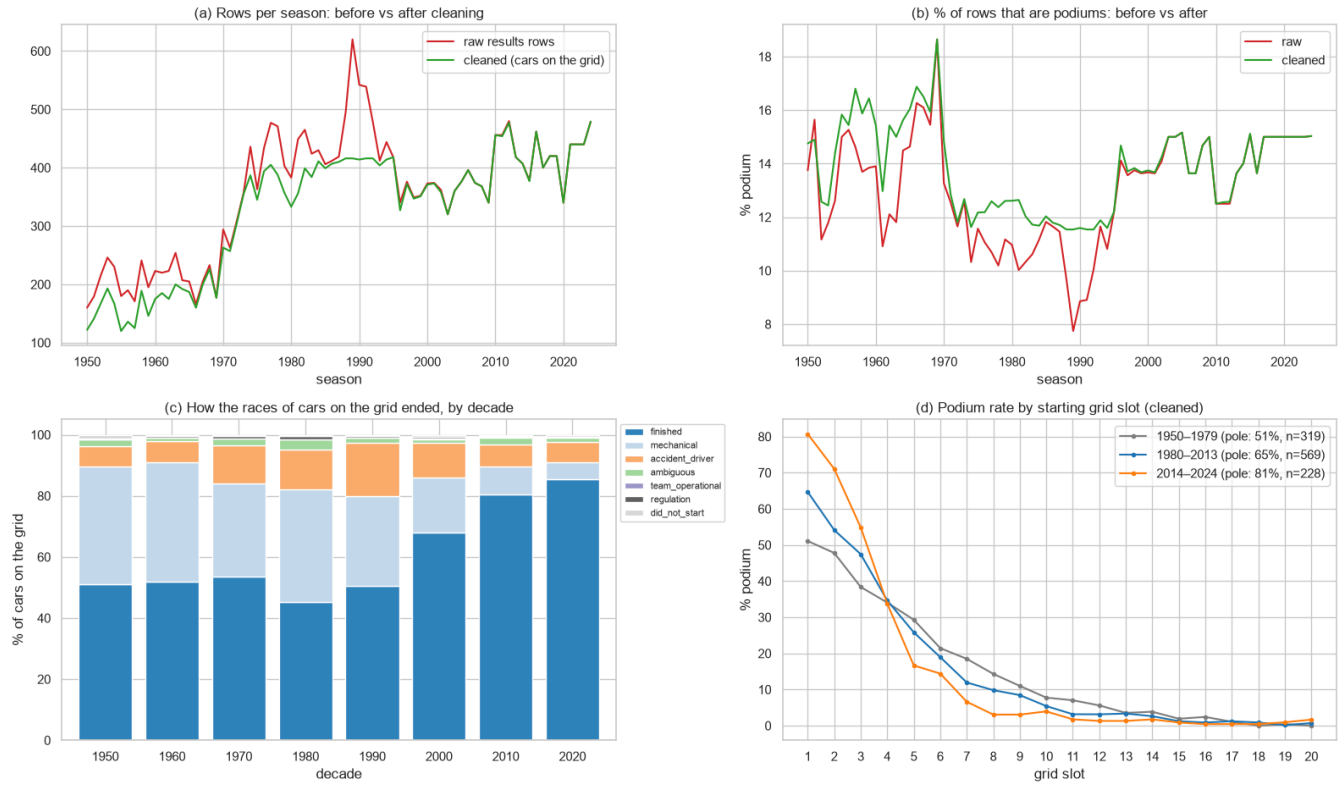


Fig 06. Before vs after cleaning: rows removed per season, podium base rate, mechanical-retirement share, and podium rate by grid slot per era.

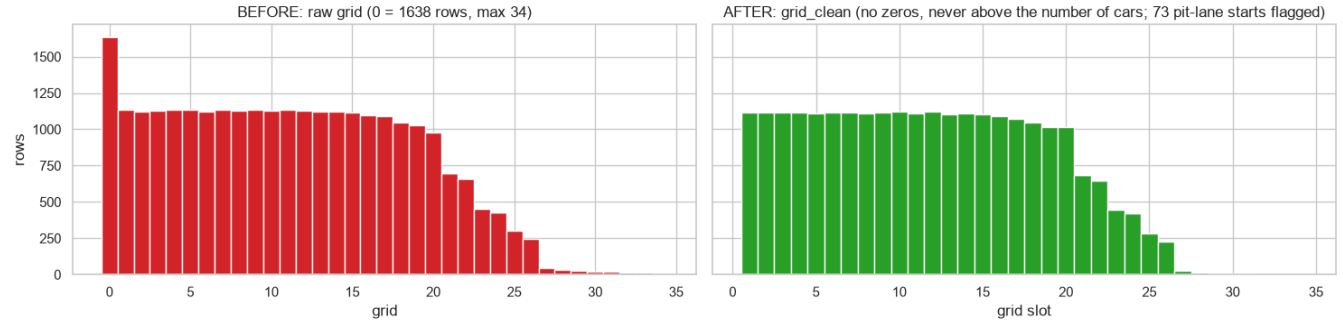


Fig 06b. Starting-position column before and after the grid recoding (D7): the spike at 0 is gone.

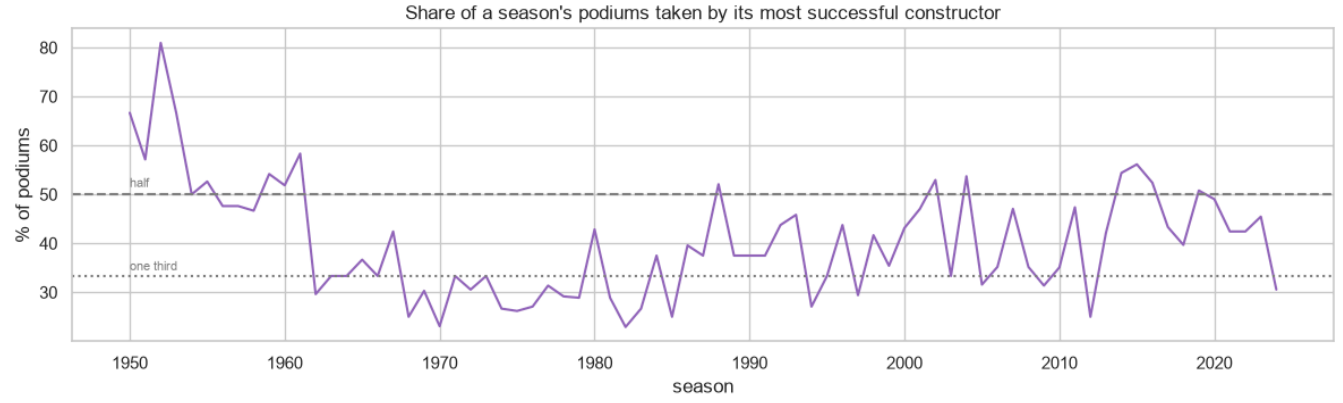


Fig 07. Share of each season's podiums taken by the top constructor (median 37.5%): the reason constructor features are included.

3. Data-engineering questions

All three questions are answered from the cleaned driver–race table, which already joins `results`, `races`, `status` and `drivers`, plus the extra tables listed for each question. Because the samples per circuit and per team are small, we report 95% Wilson intervals and a significance test with each answer.

Q1. Which circuits convert a front-row start into a podium most often, and does that change after 2014?

Front-row starts have converted into podiums more often since 2014: the pooled rate rose from 68.0% (n = 400) in 2003–2013 to 75.9% (n = 456) in 2014–2024, which is significant (chi-square p = 0.013). Individual circuits, however, cannot be separated statistically.

- **Best circuits by grid:** 2014–2024 Suzuka 89%, Sepang 88% (8 starts), Yas Marina 86%, Monaco 85%; Hungaroring last (59%). In 2003–2013 Monaco and Monza led (77%), then Silverstone (73%).
- **Does the ranking change?** Some circuits move a lot: Monza went from joint 1st to 12th, and Yas Marina from 13th to 3rd. But the rank correlation is weak and not significant (Spearman ρ = 0.21, p = 0.42). Circuits are indistinguishable within each period (chi-square p = 0.99 and 0.80). Monza's fall is a single podium fewer (17/22 → 16/22).
- **Qualifying vs grid matters.** Drivers who qualified *and* started on the front row podiumed **72.5%** of the time (n = 819). Front-row qualifiers sent back by a penalty podiumed only **28.1%** (n = 32), and drivers promoted by others' penalties 62.5% (n = 32).

Tables: `results` ⋈ `races` ⋈ `circuits` ⋈ `qualifying` ⋈ `sprint_results`. *Grain:* driver–race, aggregated per circuit and period; circuits with ≥ 8 front-row starts in both periods (17 circuits). *Missingness:* 22 of 8,964 rows from 2003–2024 lack a qualifying position; all started 19th or worse, so none is a front-row case. *Cutoff:* grid, qualifying and sprint are all pre-race. *(figs 08, 08a)*

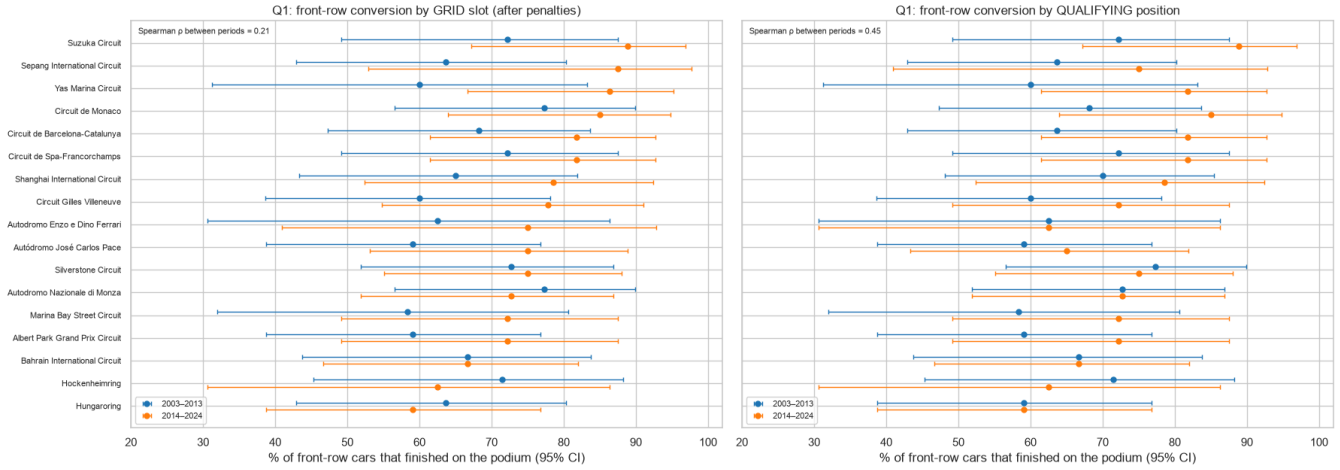


Fig 08. Q1: front-row-to-podium conversion per circuit and period, with 95% Wilson intervals.

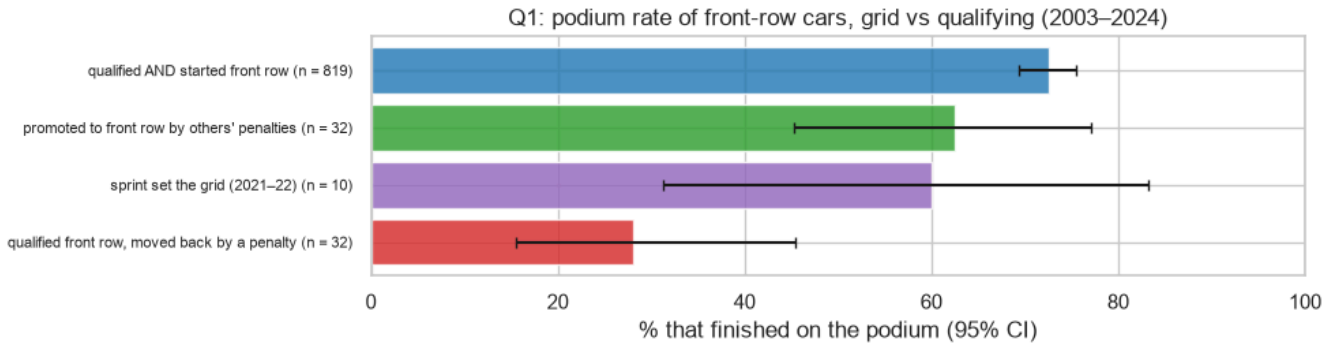


Fig 08a. Q1: podium rate of front-row qualifiers who started there, were promoted, or were moved back by a penalty.

Q2. Do drivers podium more often at home, controlling for starting position?

We find no home advantage. In the raw data home drivers podium 2.1 percentage points less often than other drivers. Comparing drivers within the same grid bucket and era, the gap shrinks to −0.8 points, and a logistic regression of podium on home, grid slot and era (standard errors clustered by race) gives an odds ratio of 0.93 (95% CI 0.79–1.11, p = 0.43).

- The raw gap is a composition effect: 13.1% of home starts come from one-off local entrants (⩽ 5 career starts) against 1.4% of away starts.
- The 15 strata of the grid bucket × era heatmap range from −15.0 to +7.7 pp, with both signs in every era.
- This is an association, not a causal effect.

Tables: `results` ▣ `races` ▣ `circuits (country)` ▣ `drivers (nationality)` with an explicit nationality→country map (D10). *Grain:* driver–race; 1,944 home starts of 24,715. *Missingness:* none; unmapped nationalities count as away. *Cutoff:* all inputs are known pre-race. *(fig 09)*

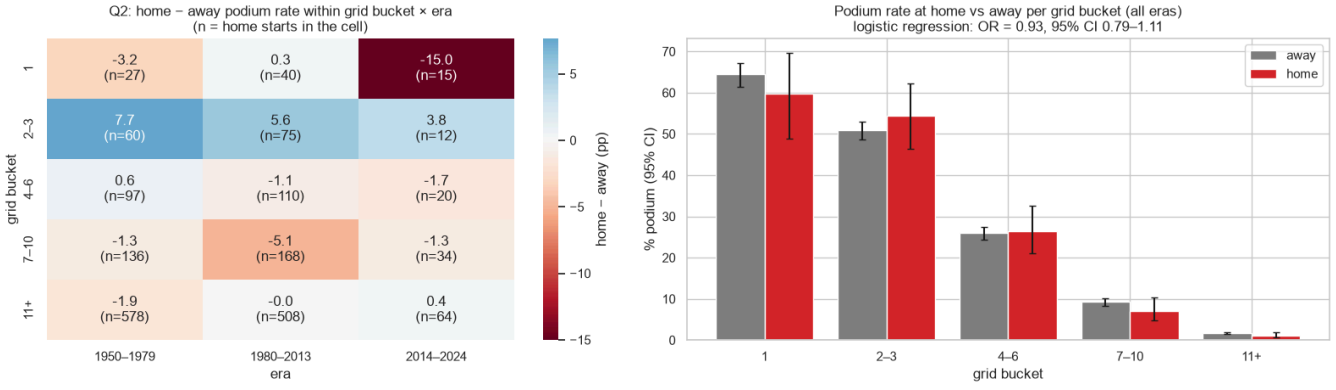


Fig 09. Q2: home − away podium rate in each grid bucket × era stratum (left) and home vs away podium rate per grid bucket (right).

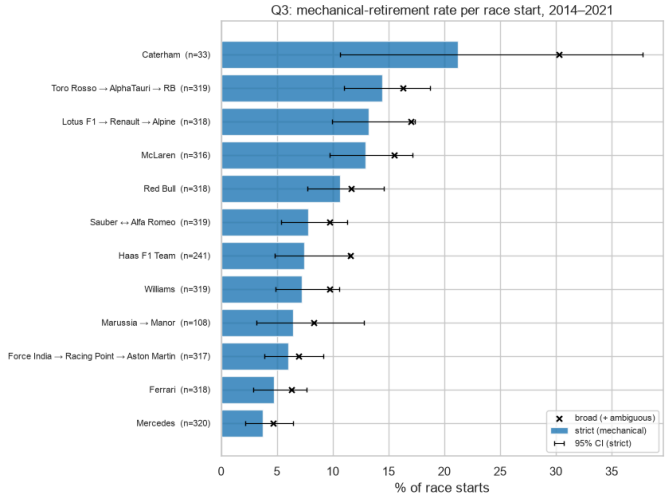
Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Mechanical retirements became less common overall, falling from 8.9% of race starts in 2014–2021 to 5.5% in 2022–2024. The teams differ clearly in the first period but not in the second.

Period	Highest rates (strict definition)	Lowest	Teams differ?
2014–2021	Caterham 21.2% (n = 33), Toro Rosso → AlphaTauri → RB 14.4%, Lotus F1 → Renault → Alpine 13.2%, McLaren 13.0%	Mercedes 3.8%, Ferrari 4.7%	yes, p < 0.001
2022–2024	Lotus F1 → Renault → Alpine 10.4% (CI 6–17%), Sauber ↔ Alfa Romeo 8.8%, Toro Rosso → AlphaTauri → RB 5.9%	McLaren, Red Bull, Aston Martin lineage 3.7% each	no, p = 0.18

- McLaren (13.0% → 3.7%) and Red Bull (10.7% → 3.7%) improved most; only Sauber ↔ Alfa Romeo and Mercedes rose slightly.
- Team order does not carry over (ρ = 0.27, p = 0.45, 10 lineages).
- Adding ambiguous labels ("Retired", "Puncture") keeps the same top teams in both periods.

Tables: `results` `races` `status` `constructors`. *Definition:* mechanical = the status names a failed car component; crashes, driver causes, regulation decisions and team pit errors are excluded (D3). *Grain:* one row per car that took the start, grouped by team **lineage** so renames (e.g. Renault → Alpine in 2021) are not split across the period boundary; lineages with ≥ 30 starts. *Missingness:* none in status. *Cutoff:* status is post-race, so it is only used here, never as a feature for the current race. *(fig 10)*



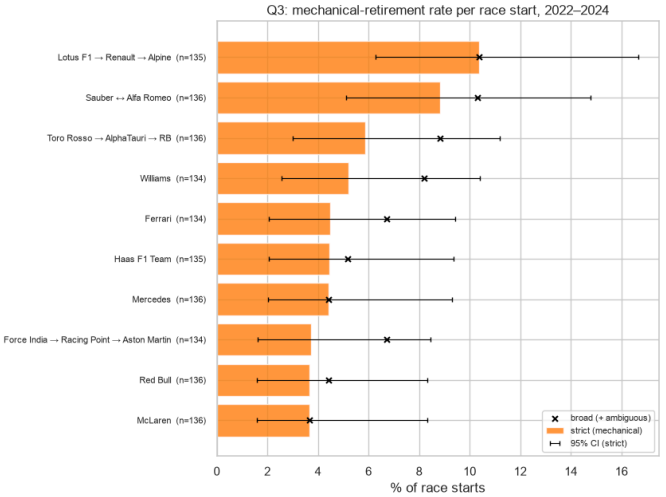


Fig 10. Q3: strict mechanical-retirement rate per team lineage with 95% intervals; × = broad rate; n = race starts.

4. Selected features and why each is valid at the cutoff

The models use 18 features in five groups, selected from 21 candidates. Each of them is known after qualifying and before the start. We never use anything from the race being predicted (result, points, laps, status, pit stops, lap times) or the standings published after it.

Group	Features	Why it is known pre-race	Evidence (standalone ROC-AUC, training seasons)
Grid	`grid_pos` (re-ranked slot after penalties), `grid_pct` (slot ÷ field size)	the grid is fixed after qualifying and penalties	0.883 / 0.880; the strongest single signal
Qualifying	`quali_position`, `quali_gap_pct` (Q1 lap vs fastest Q1, %), `grid_penalty` (grid − qualifying; vs sprint result on 2021–22 sprint weekends)	qualifying and sprints run before the race; Q1 is the only session every driver runs	0.905 / 0.830 / 0.543; penalised front-row qualifiers podium 28% vs 73% (Q1)
Standings	driver and constructor points share of leader, rank, win share	taken from the table published **after the previous race**; at a season opener, the previous season's final table (D13)	0.72–0.83; constructor ≥ driver
Form	driver and constructor podium rate and average finish over the last 5 races; constructor mechanical-failure rate over the last 10	rolling windows after `shift(1)`, so the current race is never in its own window (D14)	0.79–0.83; mechanical rate 0.618
Context	`driver_age`, `experience` (earlier races on the grid)	date of birth and race count are known in advance	0.555 / 0.663

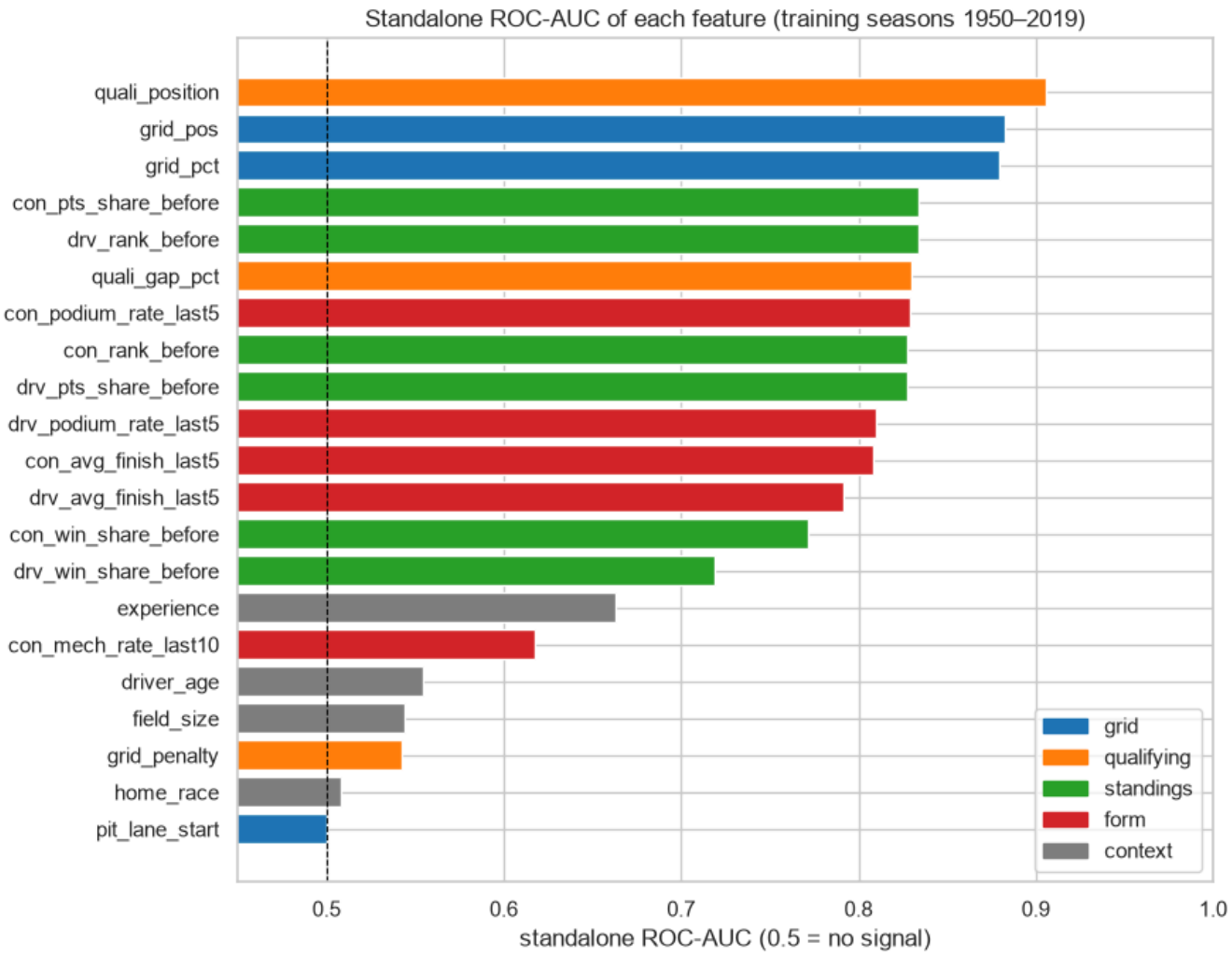


Fig 11. Standalone ROC-AUC of each candidate feature on the training seasons (0.5 = no signal).

We checked the leakage rules in the notebook. Before the 2023 Belgian GP Verstappen had 281 points and after it 314; our feature uses 281. A form value recomputed by hand from earlier races matches the feature (0.40), and an assertion confirms that every standings row comes from an earlier race.

The main pieces of evidence behind the feature choices were:

- On the same rows, grid and qualifying position are practically equal (ROC-AUC 0.904 vs 0.905). Both are kept, because they differ exactly in the penalty cases Q1 showed matter.
- One constructor took over half of all podiums in 15 seasons (median top-team share 37.5%), so constructor standings and form are included, not just driver features.
- Shares instead of raw points and wins, because points per win (8 → 25) and season length (7 → 24 races) changed.
- Raw qualifying lap times are not used: a Monaco lap (~70 s) and a Spa lap (~105 s) are not comparable.

- Sprint results are not features (D12): sprints only exist from 2021, so no training season contains them.

Three candidates were left out of the models (D30): `pit_lane_start` (0.501), `home_race` (0.508) and `field_size` (0.544). They show no signal on their own, and their information is already elsewhere: a pit-lane start is the last grid slot, the field size is part of `grid_pct`, and Q2 found no home effect. In the ablation we add each of them back to confirm that dropping them costs nothing (Section 7).

Some features have gaps for historical reasons (fig 13b). The qualifying position is missing for all training rows before 1994, for 53.6% in 1994–2002 and for 0.3% in 2003–2019. Constructor standings do not exist before 1958, and form is missing at a driver's first race. Since these gaps are structural, we fill and flag them in the model pipeline instead of dropping the rows (Section 5).

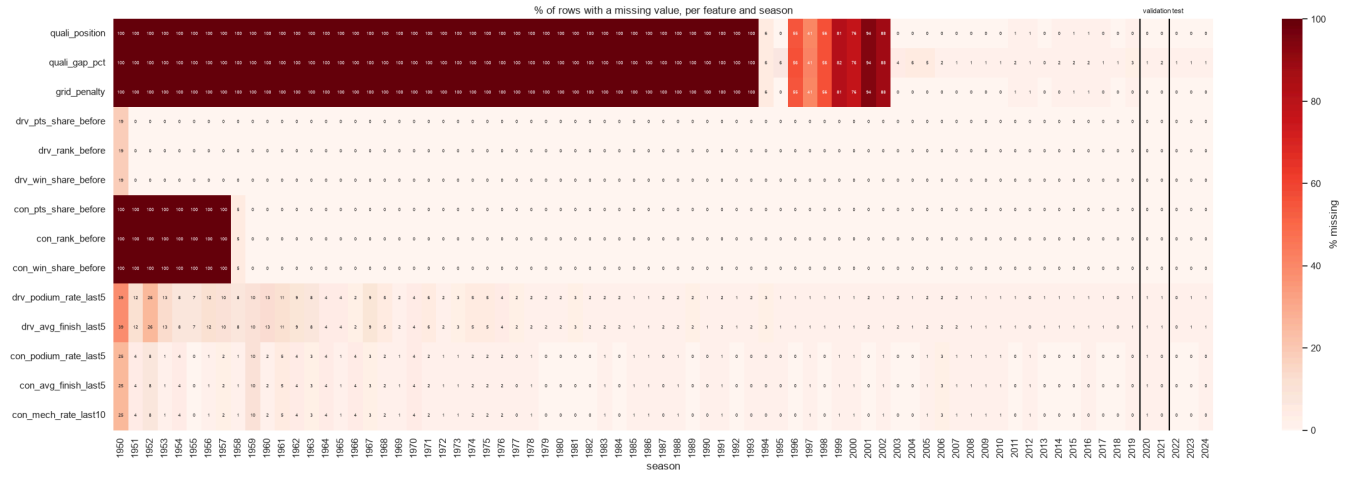


Fig 13b. Share of rows without a value per season, for each feature with gaps.

5. Split, preprocessing, models and metrics

We split the data by season rather than by random rows (D18):

Split	Seasons	Races / rows	Podium rate	Used for
Train	1950–2019	1,007 / 22,576	13.4%	fitting models and preprocessing
Validation	2020–2021	39 / 780	15.0%	early stopping, hyperparameters, threshold, model choice
Test	2022–2024	68 / 1,359	15.0%	reported once; never used for a choice

A random split would put drivers from the same race in train and test, and would train on 2023 to predict 2022. The time split copies real use: predicting the next race from past races. The test seasons start with the 2022 ground-effect rule change, so they also test transfer to new regulations. The training window starts in 1950 because no later start year scored better on validation (Section 7).

For preprocessing (D19), missing values are filled with the training median, and for each of the 14 features that have gaps we add an indicator column marking the filled values, giving 32 columns in total. The logistic regression and the neural network are also standardised with `StandardScaler`; the tree models do not need this. All steps are fitted on the training seasons only and then applied unchanged to validation and test.

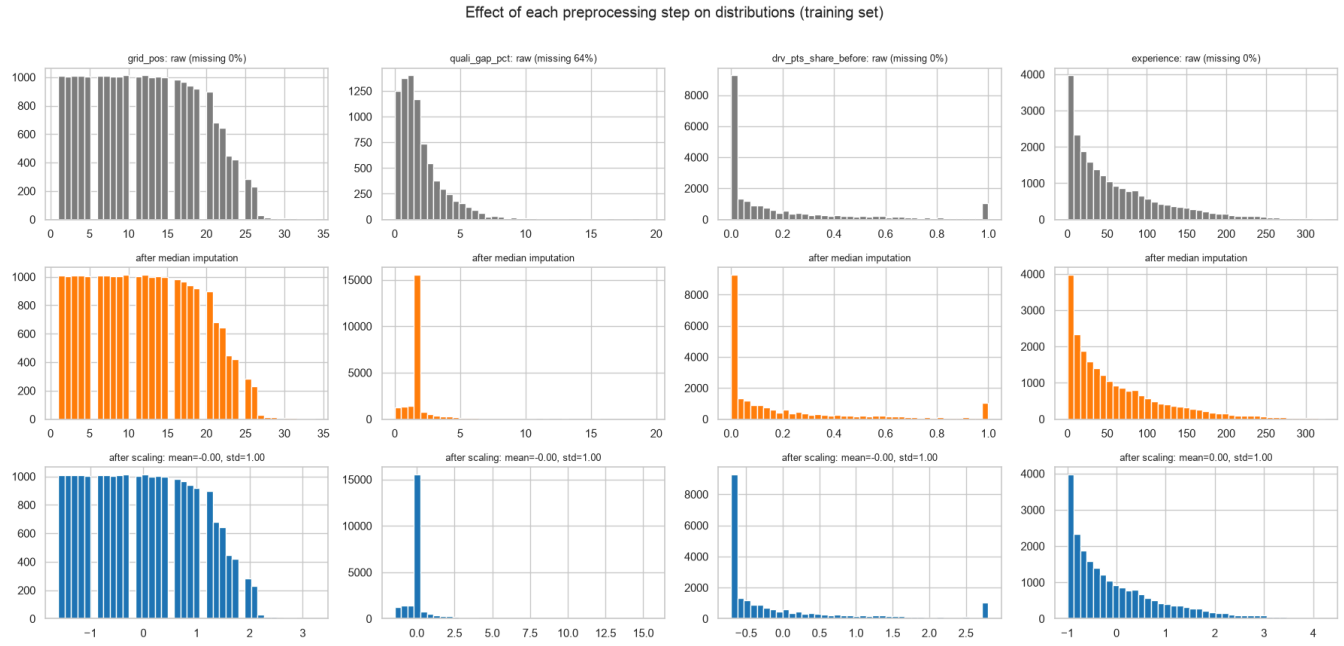


Fig 15. Effect of preprocessing on the feature distributions: imputation adds a spike at the median (hence the indicators); scaling centres each feature at 0 with SD 1.

All models are trained on the same rows with the same 18 features:

#	Model	Role	Settings
0	Grid rule, score = 1 / grid slot	reference, no learning	—
1	Logistic regression	**baseline**	L2, C = 1
2	Random forest	statistical model	400 trees, min leaf 100, chosen on validation; class weights gave no gain
3	Gradient boosting (default)	statistical model	lr 0.1, 100 trees, 31 leaves

#	Model	Role	Settings
4	Gradient boosting (tuned)	hyperparameter attempt	best of 54 combinations on validation: lr 0.01, 50 trees, 7 leaves
5	Shallow neural network (Keras)	neural network	32 inputs → 32 → 16 ReLU → sigmoid; dropout 0.2, L2 1e-4, Adam 1e-3, batch 128, early stopping on validation PR-AUC (patience 15); 1,601 weights

No model uses class weights (D20). Weights would push every probability up, and the inference function must return a usable probability. The imbalance is handled by the threshold and by using PR-AUC.

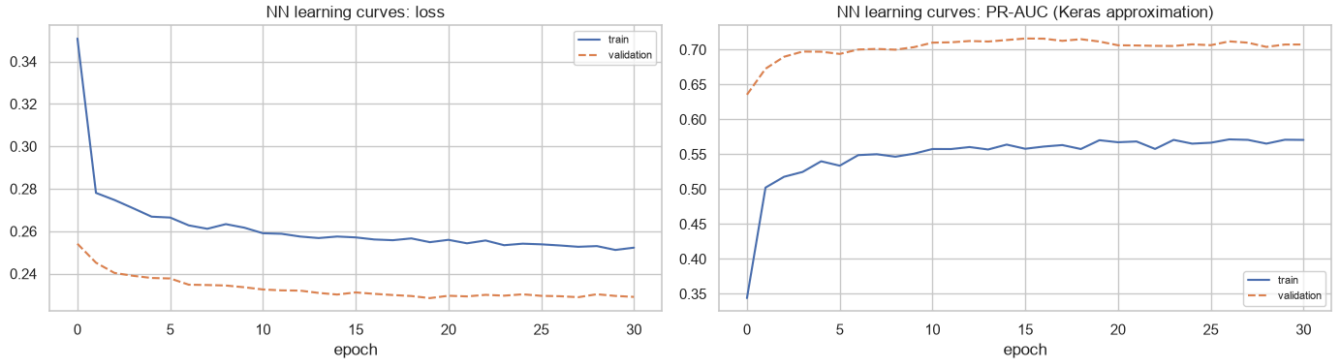


Fig 17. Shallow network training: loss and PR-AUC per epoch on train and validation; early stopping keeps the best epoch.

For evaluation (D21–D23) we use the following metrics:

- **Primary: PR-AUC.** Only ~15% of drivers podium. PR-AUC scores how clean the top of the ranking is; a random model gets 0.15 here instead of ROC-AUC's 0.5, so weak models are exposed.
- **Secondary: ROC-AUC**, for overall separation.
- **F1, precision, recall** at the threshold that maximises F1 on **validation**; the same threshold is applied to train and test.
- **Brier score** checks calibration. Accuracy is not used: always saying "no podium" scores 0.850.
- **Uncertainty:** 95% bootstrap intervals over races (1,000 resamples). A difference whose interval includes 0 is treated as no evidence of a difference.

6. Model comparison

All learned models beat the grid rule clearly, but none of them beats the logistic-regression baseline by more than the uncertainty, so we recommend the logistic regression (D24). On validation, its gain over the grid rule is 0.031–0.144 PR-AUC (95% CI), which lies entirely above zero.

Validation (2020–2021, 39 races). Threshold chosen here.

Model	PR-AUC (95% CI)	ROC-AUC	F1	Precision	Recall	Brier	Threshold
0. Grid rule	0.639 (0.541–0.745)	0.910	0.692	0.692	0.692	0.083	0.333
1. Logistic regression (baseline)	0.723 (0.612–0.836)	0.932	0.699	0.613	0.812	0.067	0.241
2. Random forest	0.733 (0.624–0.840)	0.932	0.708	0.643	0.786	0.067	0.279
3. Gradient boosting (default)	0.724 (0.627–0.820)	0.924	0.693	0.702	0.684	0.070	0.398
4. Gradient boosting (tuned)	0.736 (0.634–0.834)	0.926	0.704	0.669	0.744	0.094	0.212
5. Shallow NN (32-16)	0.716 (0.607–0.827)	0.929	0.703	0.641	0.778	0.068	0.265

Test (2022–2024, 68 races, unseen). Same models and thresholds.

Model	PR-AUC (95% CI)	ROC-AUC	F1	Precision	Recall	Brier
0. Grid rule	0.645 (0.590–0.708)	0.913	0.647	0.647	0.647	0.081
1. Logistic regression (baseline)	**0.722** (0.658–0.785)	0.937	0.673	0.546	0.877	0.068
2. Random forest	**0.722** (0.664–0.779)	0.940	0.672	0.563	0.833	0.069
3. Gradient boosting (default)	0.693 (0.634–0.754)	0.930	0.646	0.667	0.627	0.071
4. Gradient boosting (tuned)	0.696 (0.635–0.759)	0.936	0.680	0.622	0.750	0.096
5. Shallow NN (32-16)	0.711 (0.651–0.769)	0.937	0.667	0.569	0.804	0.070

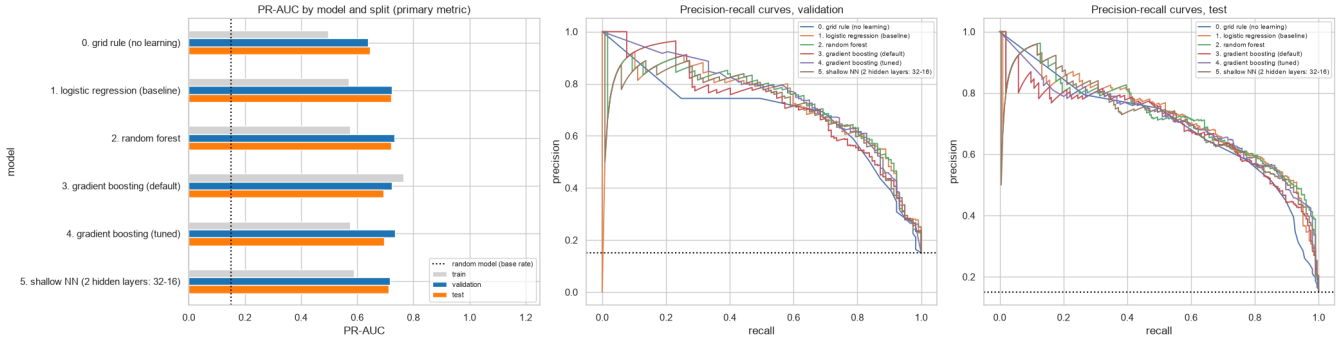


Fig 18. PR-AUC per model on train, validation and test (dotted line = random model), and precision–recall curves on validation and test.

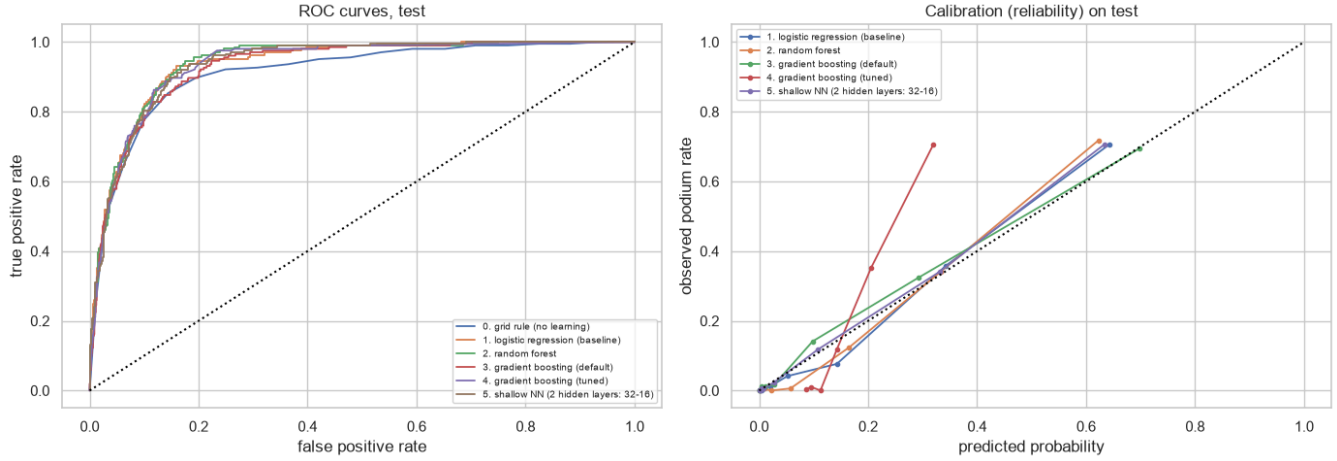


Fig 19. ROC curves and calibration of every model (calibration drawn on test, for reference only).

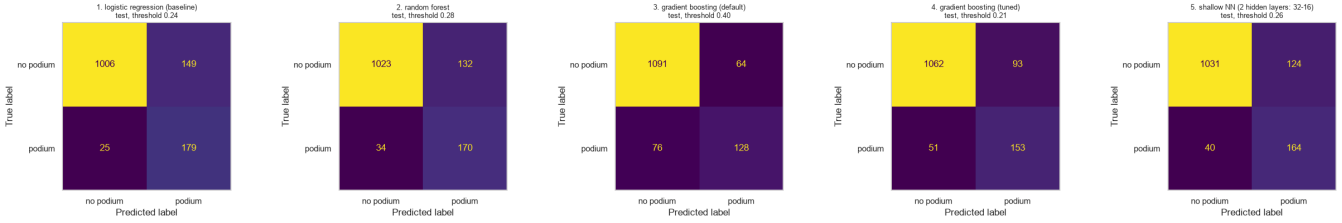


Fig 19b. Confusion matrices on test at each model's validation-chosen threshold.

Train (1950–2019). PR-AUC / ROC-AUC / F1: grid rule 0.496 / 0.883 / 0.543 · logistic regression 0.570 / 0.900 / 0.582 · random forest (out-of-bag) 0.573 / 0.901 / 0.585 · gradient boosting default 0.765 / 0.959 / 0.686 · tuned 0.574 / 0.899 / 0.588 · network 0.588 / 0.906 / 0.588.

Several points stand out in these results:

- **Differences between learned models are noise.** Ranked by validation PR-AUC, using the 3-seed mean for the forest and the network: tuned GB 0.7361, RF 0.7276, default GB 0.7235, baseline 0.7226, NN 0.7190. Every *model − baseline* interval includes 0 (tuned GB: −0.023 to +0.040). The seed spread alone is 0.7158–0.7207 for the network and 0.7230–0.7334 for the forest.

- **Overfitting.** Default gradient boosting scores 0.765 on train but 0.724 on validation, and 0.693 on test. Tuning removes the gap (0.574 train), but the best of 54 settings is optimistic on 39 races: on test, tuned and default score about the same.
- **Calibration.** Brier scores of 0.067–0.070 mean the outputs of LR, RF, NN and default GB can be read as probabilities. Tuned GB (0.094) learns so little that its outputs stay near the average.
- **Train is lower than validation, even for the grid rule** (0.496 vs 0.639). The pre-1994 seasons are much harder to rank (grid rule 0.39–0.46 per period vs ~0.69 in 2014–2019), and the podium rate is lower in training. This is not leakage.
- **Thresholds are low** (0.241 for the baseline), so the baseline predicts about 4.0 podiums per race on validation and 4.8 on test, more than the 3 that are possible. Each driver is scored independently (Section 9).
- **The network has no advantage** on ~22,600 tabular rows, which is consistent with all models relying on the same strong signals.

We chose the logistic regression because it ties for the best test PR-AUC, gives the same result on every run, is well calibrated (test Brier score 0.068) and has coefficients that can be read directly. The more complex models did not show that their extra complexity pays off. Its main weakness is that each feature changes the log-odds linearly, so interactions such as pole position mattering more at Monaco are only captured if we build them as features.

7. Experiments

7.1 Preprocessing-order experiments

For these choices correctness comes first, because a wrong order can score just as well as the right one when it uses information it should not have. Scores are validation PR-AUC of the logistic regression unless stated otherwise.

Experiment	Compared orders	Validation PR-AUC	Decision
E1	split → fit preprocessing **vs** fit on all rows → split	0.7226 vs 0.7230	split first. The leaky version's medians already contain 2020–2024 (e.g. `quali_gap_pct` median 1.500 → 1.343)

Experiment	Compared orders	Validation PR-AUC	Decision
E2	impute → scale **vs** scale → impute	LR 0.7226 vs 0.7214; NN 0.7158 vs 0.7227 (test reverses: 0.7112 vs 0.7034)	impute → scale. Gaps are within the network's seed spread
E3	derive `grid_penalty` → impute **vs** impute `quali_position` → derive	0.7226 vs 0.7208	derive first. The other order invents a penalty for 14,228 training rows
E4	fill missing qualifying with median + indicator **vs** median only **vs** grid fallback	LR 0.7226 / 0.7103 / 0.7154; GB 0.7235 / 0.7017 / 0.6633	median + indicator. Both alternatives are significantly worse for the LR (and grid fallback for GB)

7.2 Training-window check

Training from 1950 works best: no later start year improved the validation score. The table shows the mean validation PR-AUC of the logistic regression and default gradient boosting when trained up to 2019 from each start year, compared with 1950 using a paired race bootstrap.

Start year	Training rows	Mean PR-AUC	Gain vs 1950 (95% CI)
1950	22,576	0.7230	—
1960	21,070	0.7030	−0.020 (−0.033 to −0.002)
1970	19,193	0.7157	−0.007 (−0.022 to +0.005)
1980	15,737	0.7137	−0.009 (−0.029 to +0.010)
1994	10,156	0.6965	−0.027 (−0.044 to −0.010)
2003	6,825	0.6955	−0.028 (−0.049 to −0.006)

The logistic regression alone gains slightly from later starts (0.7331 from 2003), but gradient boosting loses a lot (0.7235 → 0.6579): flexible models need many rows. Earlier validation periods (2016–17, 2018–19) show no start year that is best everywhere, and 2003 is always lowest.

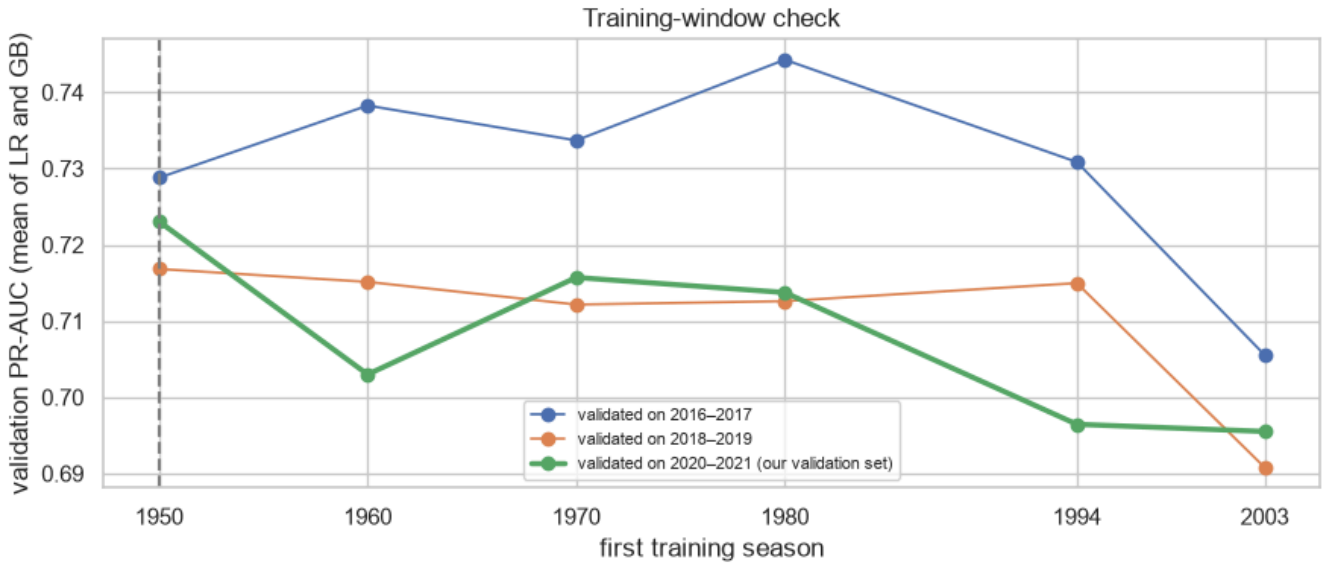


Fig 14. Training-window check: validation PR-AUC of LR and default GB by first training season.

7.3 Feature-group ablation

In the ablation we remove one feature group at a time and retrain each model with its own settings; random forest and network results are averaged over three seeds. The table shows the change in validation PR-AUC compared with using all features, and * marks changes whose 95% race-bootstrap interval excludes zero. Starting position is clearly the most important group for every model, while the other groups mainly add information together and can partly replace each other.

Variant	Logistic regression	Random forest	GB default	GB tuned	Shallow NN
All features (PR-AUC)	0.7226	0.7299	0.7235	0.7361	0.7226
Without grid	−0.003	**−0.039 ***	+0.007	−0.035	+0.004
Without qualifying	−0.015	−0.004	**−0.045 ***	−0.014	+0.004
Without standings	−0.009	−0.016	**−0.054 ***	**−0.029 ***	−0.004
Without form	+0.015	+0.004	+0.008	−0.003	+0.006
Without context	**−0.008 ***	−0.004	−0.030	0.000	−0.003
Grid group only	**−0.084 ***	**−0.091 ***	**−0.085 ***	**−0.100 ***	**−0.084 ***

- **Grid only** loses 0.084–0.100 on validation and 0.050–0.079 on test for every model, all significant. Qualifying, standings, form and context together add a lot on top of starting position.
- **Single-group removals mostly change little**, because groups overlap (grid ↔ qualifying, standings ↔ form). The logistic regression loses almost nothing without grid on validation (−0.003) but significantly on test (−0.020): in recent seasons every

driver has a qualifying position, so a linear model can switch to it.

- **Form shows no measurable contribution as a group**, since standings carry similar information.
- **Single weak features:** removing `experience` hurts the LR significantly (−0.009); removing `grid_penalty` or `driver_age` is within noise. Adding back `pit_lane_start`, `home_race`, `field_size` or all three never helps significantly (at most ±0.019).
- **No removal improves any model significantly**, so all five models keep all 18 features (D30).

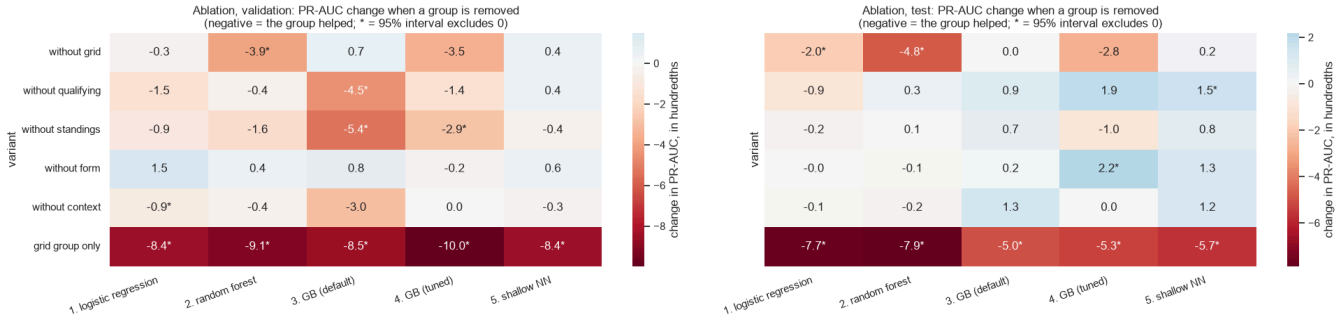


Fig 20. Feature-group ablation: change in PR-AUC per model and variant, validation and test.

8. Explainability (XAI)

All five models rely on starting position first. Beyond that, the tree models depend more on standings, whereas the logistic regression and the network make more use of qualifying. Every explanation is computed on the 2022–2024 test seasons, which the models never saw during training.

> **These explanations are not causal effects.** A feature being important means the *model relies on it to predict*, not that it *causes* podiums. For example, `con_pts_share_before` matters because strong teams both score points and finish on podiums; giving a team points would not make its cars faster. Correlated features also share importance, so one feature can look less important than the information it carries.

8.1 Global: permutation importance

Each feature, or a whole group, is shuffled across drivers 20 times, and the drop in test PR-AUC is measured. Shuffling whole groups is fairer, because correlated partners such as `grid_pos` and `grid_pct` otherwise stand in for each other *(fig 21)*.

Group shuffled	Logistic regression	Random forest	GB default	GB tuned	Shallow NN
Grid	**0.284**	**0.225**	**0.245**	**0.363**	**0.217**
Qualifying	0.060	0.001	0.009	0.006	0.038
Standings	0.044	0.076	0.054	0.049	0.055
Form	0.041	0.025	0.040	0.033	0.041
Context	0.002	0.002	−0.009	0.000	0.000

Tree models hardly use qualifying: it is missing in 63% of their training rows, so they split on grid position, which carries the same information in every season. For single features, the LR depends on `grid_pos` (0.279) and `quali_position` (0.053); the network on `grid_pct` (0.093) and `grid_pos` (0.056).

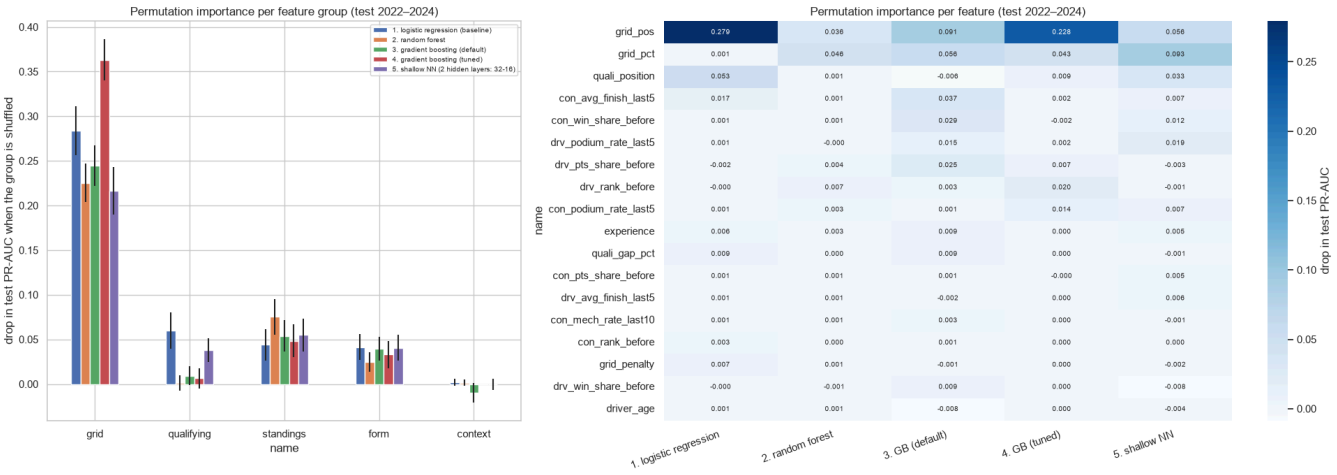


Fig 21. Permutation importance on test: drop in PR-AUC when a feature group or single feature is shuffled (20 repeats).

8.2 Global: SHAP

SHAP (model-agnostic `PermutationExplainer`, 100 background rows from 2003–2019, 303 test rows) splits each prediction into per-feature contributions in **probability**, which add up exactly to the prediction (largest gap 1e-8) *(figs 22, 23)*.

Share of mean |SHAP|	Logistic regression	Random forest	GB default	GB tuned	Shallow NN
Grid	37.6%	38.7%	35.9%	58.8%	34.3%
Standings	16.5%	33.0%	25.2%	25.0%	21.8%
Form	20.0%	19.3%	21.0%	14.1%	27.6%
Qualifying	20.2%	8.1%	6.7%	2.1%	11.6%
Context	5.7%	0.9%	11.2%	0.0%	4.8%

The logistic-regression weights point the sensible way for the main features: a later grid slot (−1.482 log-odds per SD), a worse qualifying position (−0.333) and a worse constructor average finish (−0.205) lower the prediction; recent podiums (+0.161) and a high constructor points share (+0.130) raise it. A few small weights with the "wrong" sign come from correlated inputs splitting shared information, so only group totals should be read.

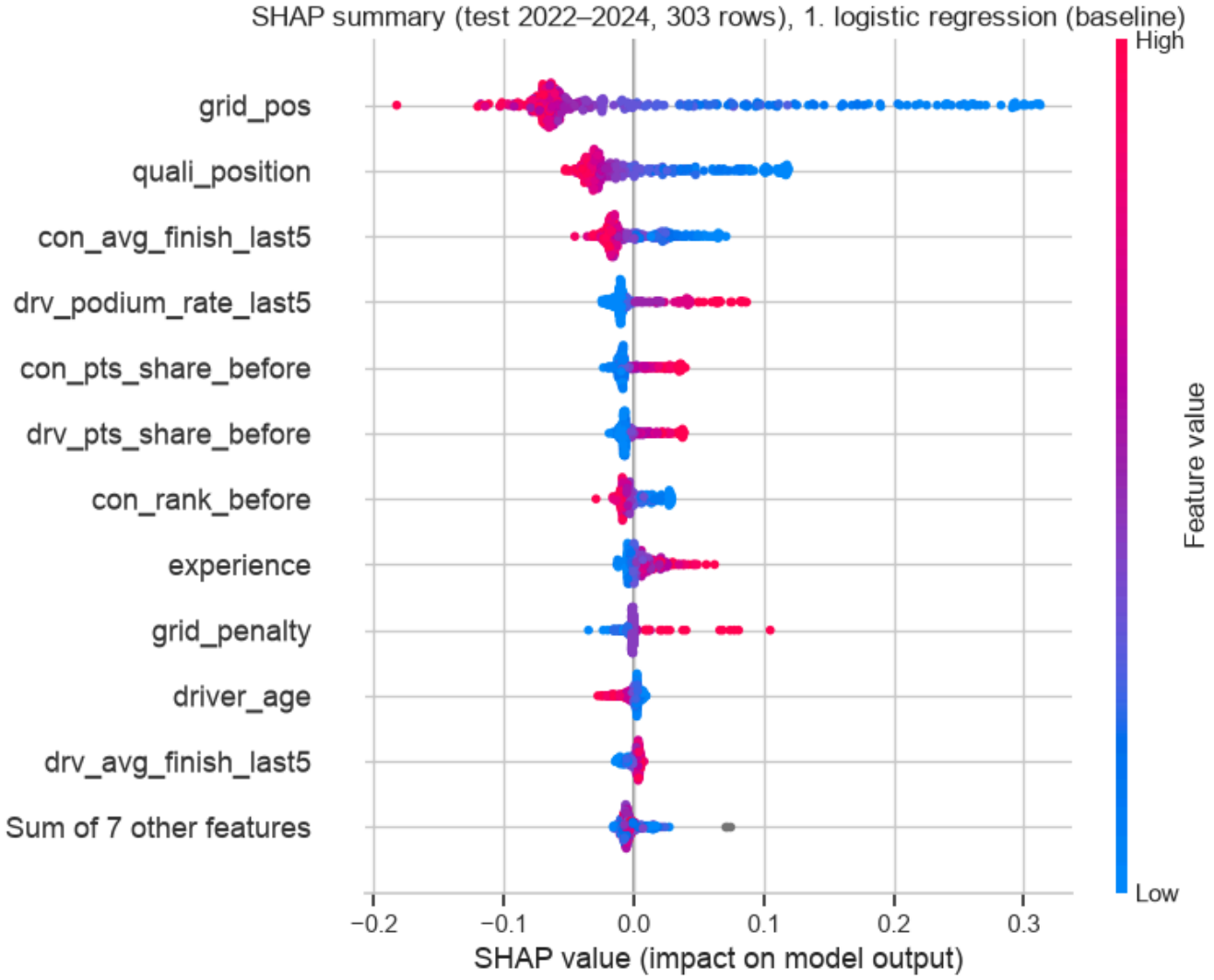


Fig 22. SHAP summary for the recommended logistic regression (303 test rows). Red = high feature value; right = pushes the podium probability up. A low (good) grid slot pushes up most.

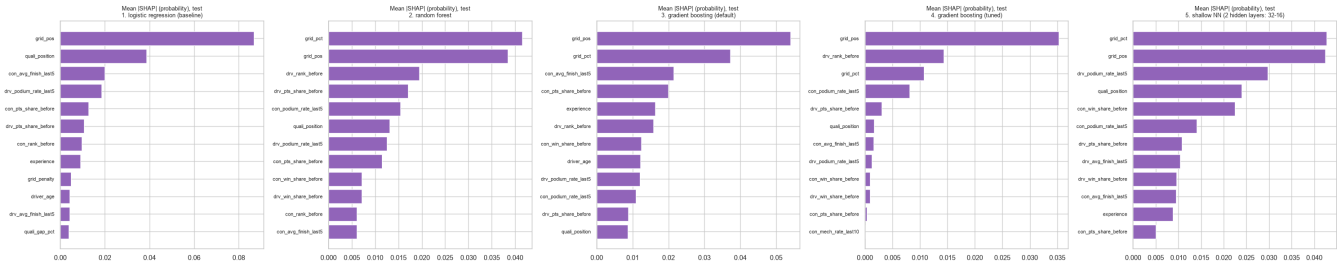


Fig 23. Mean |SHAP| per feature for all five models: grid features lead in every model.

8.3 Local: one prediction (SHAP waterfall and LIME)

As a single example we use Max Verstappen at the 2023 Belgian GP. He set the fastest qualifying time, received a five-place gearbox penalty, started sixth, and won the race.

Model	Probability	Decision	Largest SHAP contributions
Logistic regression	0.735	podium	`grid_pos` +0.118, `quali_position` +0.117, `drv_podium_rate_last5` +0.086
GB default	0.538	podium	`con_win_share_before` +0.085, `drv_podium_rate_last5` +0.058
Shallow NN	0.513	podium	`drv_podium_rate_last5` +0.114, `con_win_share_before` +0.067
Random forest	0.490	podium	`drv_rank_before` +0.050, `drv_pts_share_before` +0.045
GB tuned	0.230	podium	`grid_pos` +0.030, `drv_rank_before` +0.024

- **SHAP waterfall** *(fig 24)*. For the LR, base value 0.155 + contributions 0.580 = 0.735. The LR uses both the P6 grid slot and the pole position. The gradient-boosting models learned mostly from grid, so to them P6 looks like P6.
- **What-if (model behaviour, not causal).** Starting from P1 instead raises the LR to 0.840; from P10 it falls to 0.623.
- **LIME** *(fig 25)*. Every model's top LIME condition is `grid_pos ⩽ 6` (+0.076 to +0.131, stable in all 5 seeds), followed by standings and `grid_penalty > 0`, matching SHAP. But LIME's local linear model fits poorly (R^2 0.700 for LR, 0.34–0.52 for the others), and its own prediction for the LR is 0.279 against the real 0.735. **SHAP is the more trustworthy local explanation here** because it adds up exactly.
- **Most confident miss.** Three models (LR 0.849, RF 0.803, NN 0.801) are most confidently wrong on Verstappen at the 2024 Australian GP: pole position and championship leader, but retired with a brake failure. No pre-race feature can announce a component failure.

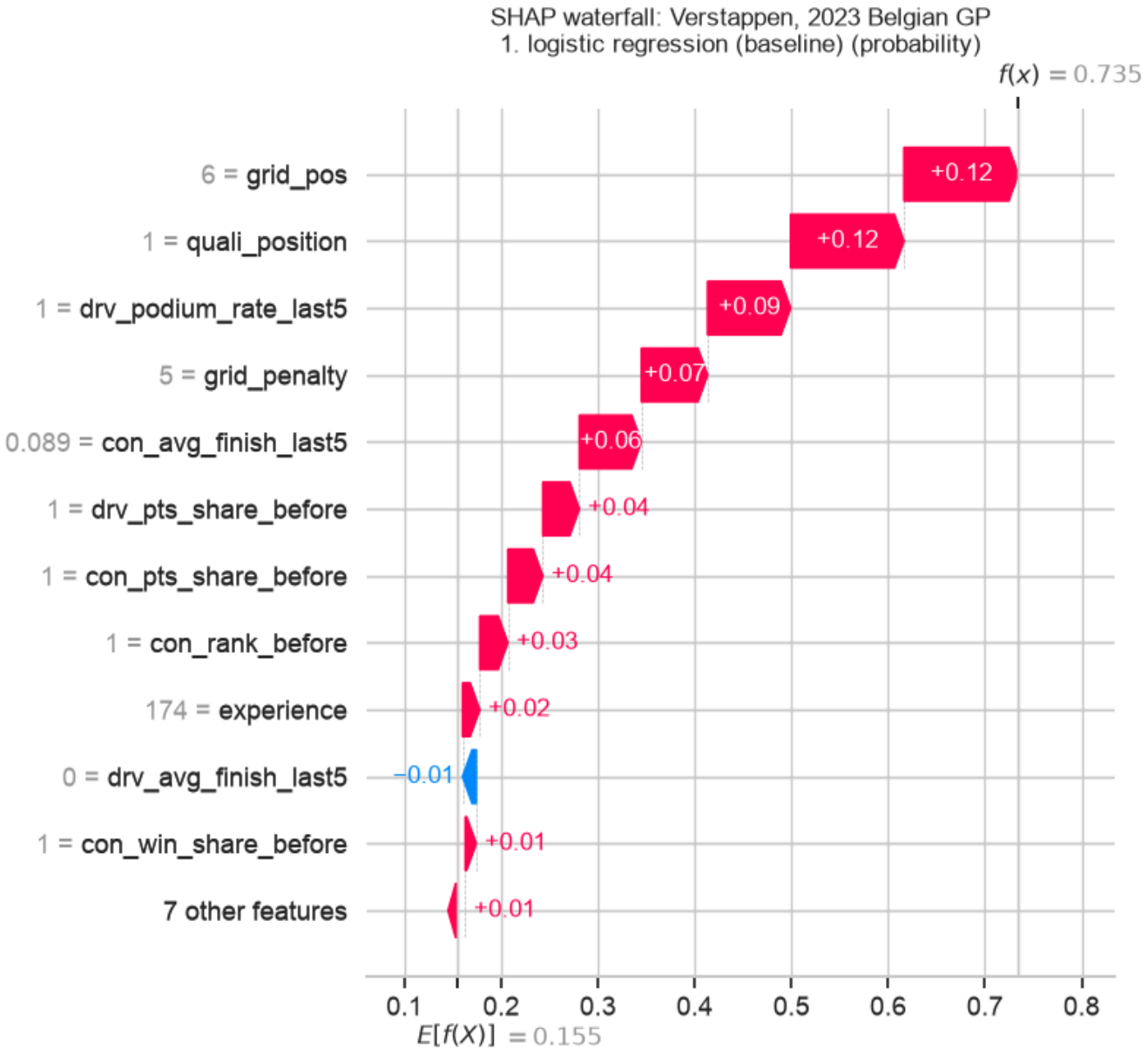


Fig 24. Local SHAP waterfall for Verstappen, 2023 Belgian GP (logistic regression): base value 0.155 plus contributions = 0.735.

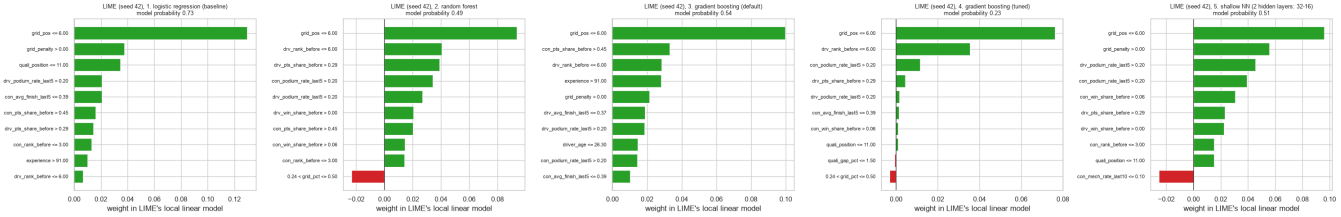


Fig 25. LIME explanation of the same prediction for each model (mean over 5 seeds).

9. Known limitations

- **Qualifying-coverage drift.** Qualifying data starts in 1994 and is patchy until 2002, so 63% of training rows have no qualifying position. The models learn qualifying features from recent seasons only; the missing-value indicators and the grid position carry the

older seasons. Pre-1994 seasons are the hardest to rank (grid rule PR-AUC 0.39–0.46 per period).

- **Era sensitivity.** Rules, reliability and field sizes changed over 75 seasons. The 2022–2024 test (new ground-effect rules) shows the model still transfers, but a future rule change may not.
- **Small evaluation sets.** 39 validation races and 68 test races give PR-AUC intervals of about ±0.06–0.11, wider than the gaps between learned models. Validation scores are slightly optimistic because validation was used for several choices.
- **Independent predictions.** Each driver is scored alone, so nothing forces exactly three podiums per race; the baseline predicts 4.0 per race on validation and 4.8 on test.
- **Missing race-day factors.** Weather, tyre strategy, safety cars, crashes and breakdowns are not in the data and cause most upsets (e.g. the most confident misses in Section 8).
- **Unusual missing-value patterns.** A record whose gaps never occurred in training gets an unreliable prediction. Removing Verstappen's qualifying and form data *raises* the network's probability (0.513 → 0.833), because that pattern only existed for pre-1994 rows.
- **Tuned gradient boosting probabilities** stay near the average (Brier 0.094); they are usable for ranking, not as literal probabilities.
- **Team identity.** A renamed constructor gets a new `constructorId`, so its form history restarts (Q3 groups lineages; the features do not).
- **Documented data quirks.** 1950s shared drives (one race still has four podium rows), vacant positions after disqualifications, broken 2024 Belgian GP race times, and pit-lane vs non-starter for raw grid 0 decided with the laps column.
- **Explanations are not causal** (Section 8).

10. Inference function and conclusion

Each model has its own inference function (`predict_podium_lr`, `_rf`, `_gb_default`, `_gb_tuned` and `_nn`). The function takes one driver's pre-race information and returns the podium probability, a yes/no decision using that model's validation threshold, and a list of the features that were missing. `grid_pos` and `field_size` are required; any other gap is filled in the same way as during training. Invalid input, such as a misspelt feature name or a grid slot outside the field, raises an error.

Record	LR	RF	GB default	GB tuned	NN	Actual
Complete: Verstappen, Spa 2023	0.735 podium	0.490 podium	0.538 podium	0.230 podium	0.513 podium	won

Record	LR	RF	GB default	GB tuned	NN	Actual
Partially missing: Guanyu Zhou, Bahrain 2022 debut (no own form)	0.004 no	0.002 no	0.001 no	0.087 no	0.001 no	no podium
Same debutant from the pit lane (raw grid 0)	0.002 no	0.001 no	0.000 no	0.087 no	0.001 no	—

In conclusion, a leakage-safe table with one row per driver per race, built from the 14 relational tables, allows podium prediction well above a grid-only rule (test PR-AUC 0.722 against 0.645). Starting position carries most of the signal, and standings, recent form and qualifying pace add further information together. The choice of model matters little on this data, so we recommend the simple and well-calibrated logistic regression. Most of the remaining errors come from race-day events such as mechanical failures and crashes, which no pre-race feature can anticipate.